

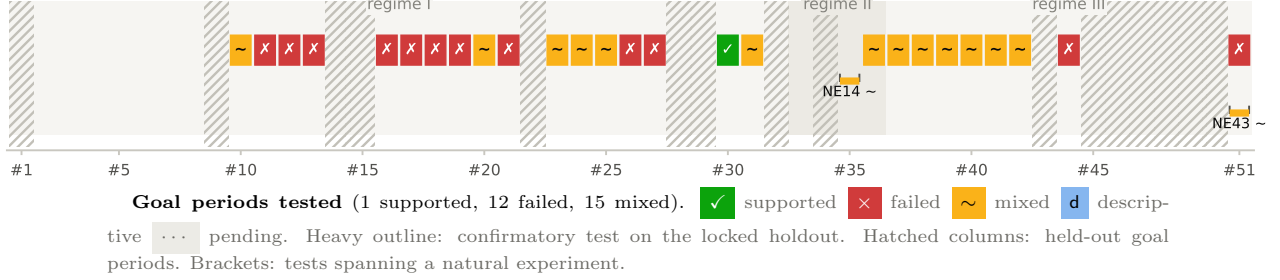
H49: Edge-trimmed regime III is a dilute ferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

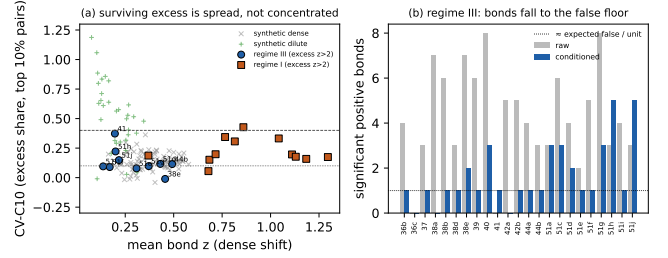
Bottom line. After the operator's schedule is trimmed, regime-III co-activation is a weak, structureless swarm-wide mode, not a dilute ferromagnet; individual bonds sit at the false-positive floor.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim is that regime-III activity coupling, with H38's day-edge drive trimmed, sits in a few strong pairs below percolation. The test infers conditioned pair couplings in 23 regime-III and 23 regime-I units. Each unit has 200 block-shift surrogates, calibrated on synthetic swarms at real schedules. **Refuted as posed.** Trimming removes 67% of regime-III bonds (17% in regime I). The remainder sits at the false-positive floor: 38 bonds against ≈ 23 expected false. The surviving collective excess (10 units, including #44 and #51) is a structureless shift of all pairs. The top 10% of pairs carry a median 11% of it, and pair structure does not replicate. #44's fine-tuning team appears in talk, not in activity. Same-lab (provider-outage) pairs carry no excess bonds. At #51's sampling the method finds strong bonds (recall 0.89). NE43: the edge drive outlives the daily bookend messages. The holdout is not run.



Observables. (a) Units with a significant edge-trimmed excess (regime III blue, regime I orange). The reference is planted swarms at village sampling (grey $\times$: dense $J = 1.2/N$; green $+$: dilute $J = 0.5$). Real units sit with the dense cloud: low CV-C10, positive mean bond z . (b) Regime-III significant bonds per unit, raw (grey) and trimmed (blue). Trimming brings them down to the ≈ 1 expected false bond per unit (dotted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not target agent pairs from 1-min activity timing. After the daily start and stop are trimmed, no small clique of agents works in lockstep: 38 bonds stand against ≈ 23 expected false. A weak common mode shared by all pairs remains. It is consistent with a fast shared field as well as with peer coupling. Monitor and steer the common mode, not individual inferred bonds. Look for pairwise structure in who talks in the same minutes: the one known team appears there (p 0.013). The corrected NE43 date is a small data contribution.

PHYSICAL MODEL

Inverse Ising (model 01). The spin $s_i(t) = \pm 1$ marks agent i active in minute t . $\delta_i(b)$ is an agent field per 30-min block. x_j is the block-centered spin; scaffold-unavailable minutes (not started, finished, consolidating, error gap) take the block mean (H38's conditioning). J_{ij} is the equal-time pseudolikelihood bond. A bond is significant when its surrogate $p < 1/n_{\text{pairs}}$. Percolation uses the Molloy–Reed κ . CV-C10 is the cross-validated share of the excess covariance ΔC in the top 10% of pairs.

$$\text{logit } P(s_i(t) = 1 \mid \cdot) = 2 \left[\delta_i(b) + \sum_{j \neq i} J_{ij} x_j(t) \right], \quad x_j = a_j (s_j - \bar{s}_j^{(b)})$$

$$z_{ij} = (J_{ij} - \langle J_{ij} \rangle_{\text{shift}}) / \sigma_{ij}, \quad \text{bond if } p(z_{ij}) < 1/n_{\text{pairs}}$$

$$\kappa = \langle k^2 \rangle / \langle k \rangle \text{ (giant component iff } \kappa > 2), \quad \text{VR} - 1 = \sum_{i \neq j} \bar{C}_{ij} / \sum_i \bar{C}_i$$

H49-dilute-ferromagnet · generated 2026-10-04 11:53 by in-
fra/summaries/build_summaries.py · card hypotheses/H49-dilute-
ferromagnet/README.md · ratings: H49 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
Synthetic: calibrated null; dilute bonds recovered	size 0.054–0.062; recall 0.47 / 0.34 / 0.89	✓ / partial
Synthetic: percolation detectable	0.13 / 0.00 / 0.77 of percolating graphs	×
P1 trimmed bonds ≤ 5% of pairs	0.95%; 38 vs ≈23 false	✓
P2 excess concentrated (CV-C10 ≥ 0.4)	0/10 units; median (dense)	0.11 ×
P3 below percolation	23/23, trivially (power)	✓
P5 trimming removes bonds in III, not I	67% vs 17%	✓
NE43 edge drive outlives bookends	×0.77, ×0.93	✓
#44 team is a bond cluster	activity p 0.26, 0 bonds; talk p 0.013	×
#51 bonds recur; same-lab (R2)	r 0.02; OR 0.50	×
NE14 bonds persist II→III	no reliable pairs either side	n/a

GOAL PERIODS IN DETAIL

G10 ~ mixed
G11 × failed
G12 × failed
G13 × failed
G16 × failed
G17 × failed
G18 × failed
G19 × failed
G20 ~ mixed
G21 × failed
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Equal-time inference does not see coupling delayed by minutes, such as read-out-gated replies (H08). The verdict covers equal-time bonds only. Dense J_0/N coupling and a fast shared field look the same here. The post hoc size law favours a field, but it relies on cross-period N . Individual bonds need $J \gtrsim 0.5$ and ≥ 5 days for detection. CV-C10 became a pooled ratio after the first pass, with no verdict change. All results are rebuilt after the activity-bins join bug; no verdict changed. The holdout is not run.

ROUND 2

Fit lagged bonds at the read-out lag. Identify the shared field (human messages, nudges, provider latency) and test the size law at roster joins within #51. Test dilute against dense in the talk channel, where the #44 team appears. Run `confirm.py`.

- **Essence.** whether an agent swarm’s collective co-activation is carried by a few identifiable pairs (targetable) or by a swarm-wide mode (steer the field, not the pairs). For regime-III activity the answer is the mode.
- **H49-R1** Lagged bonds.
- **H49-R2** Identify the shared field.
- **H49-R3** Talk bonds.
- **H49-R4** Power-corrected percolation.
- **H49-R5** Run `confirm.py` (C1–C6).

Second figure. Planted swarms at three real schedules (30 replicates each). (a) Raw inference fakes bonds from the day edges; conditioning brings the null to the floor. (b) CV-C10 separates dense from dilute when the excess is significant. (c) Every structure reads as “below percolation”, even a percolating one. The concentration call is rare for dense coupling and common for dilute coupling.

